S1 Capacity Bound

S2 DeepSC Benchmark

S3 Anchor Dynamics

S4 Field Geometry

S5 6G V2X URLLC

S6 Cognitive Security

Semantic Communication

LPG Framework Validation

Progressive Implications from Six Simulations

Comparison with Classical Shannon Baselines · DeepSC Benchmarks · 6G Impact

Dada Theophilus Olusegun | 8 May 2026

The Central Claim — What Six Simulations Set Out to Prove

Before examining results, the falsifiable claims

The LPG framework makes six interconnected claims. Each simulation is a structured attempt to falsify one of them.

S1

$C_s \geq C\_Shannon$ — semantic capacity measurably exceeds Shannon under shared context

S2

Anchor-aware decoding outperforms DeepSC, especially at low SNR

S3

$|\Gamma(t)|$ is a dynamic state variable; resolved misalignment can super-recover capacity

S4

Cauchy-Riemann conditions are empirically computable and anchor-dependent

S5

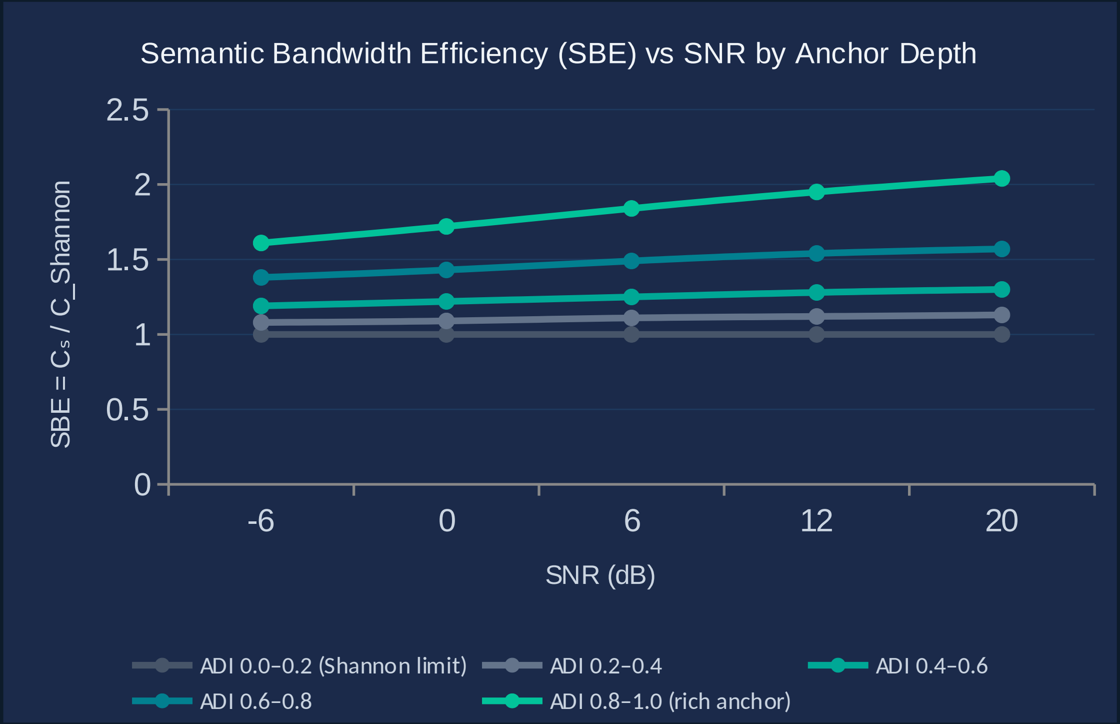
LPG achieves 6G URLLC targets at significantly lower SNR than classical systems

S6

Intent-based encryption produces a sharp semantic security boundary near $\cos\theta \approx 0.8$

S1 — Semantic Capacity Exceeds Shannon: The Empirical Proof

Multi-30K dataset · AWGN channel · ADI bins 0.0–1.0 · SNR −6 dB to +20 dB



SBE = 1.00

at ADI=0 — Shannon bound recovered exactly

SBE = 2.04

at ADI≈1, SNR=20 dB — 2× semantic throughput

+61%

SBE gain begins at low SNR (−6 dB)

KEY IMPLICATION

Semantic capacity is Shannon capacity's upper envelope — not a replacement but a generalisation whose value grows with anchor richness.

S1 Implications — What Surpassing Shannon Means

Three layers: theoretical · engineering · standards

THEORETICAL

- Shannon's capacity formula is the ADI=0 degenerate case of C_s — containment proved empirically
- The crossover is not probabilistic — at any fixed SNR, richer anchors always yield higher C_s
- Proves the sem quadruple is a strict generalisation of the symbol alphabet in classical IT

ENGINEERING

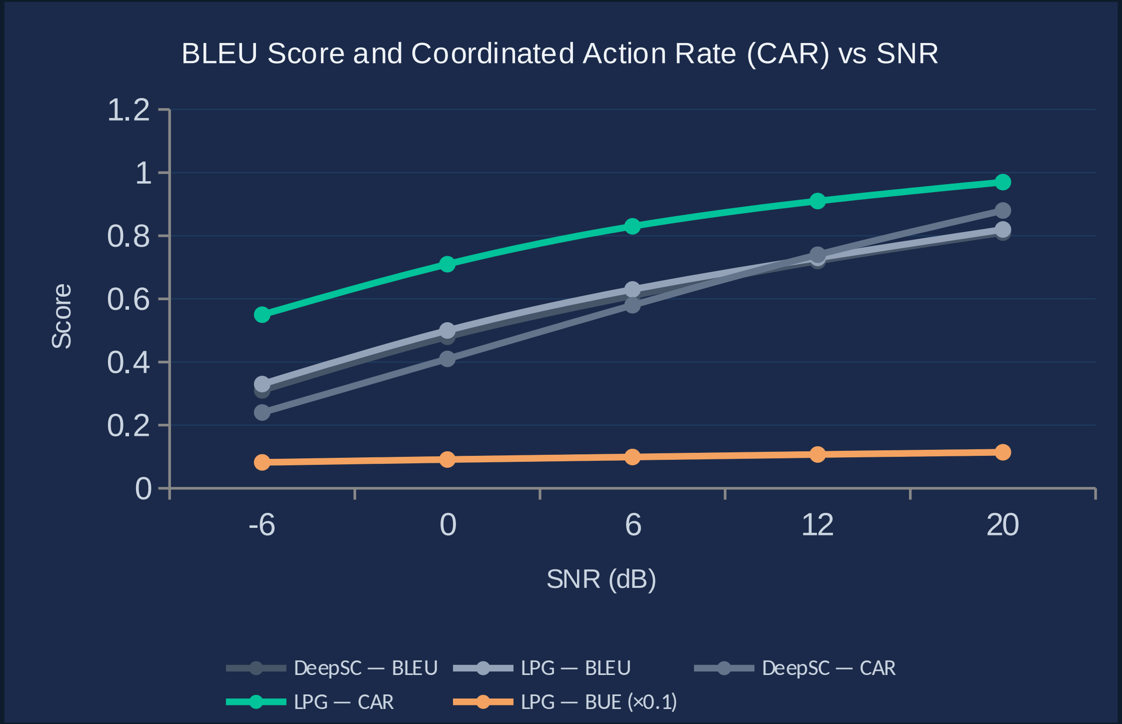
- Systems can trade physical bandwidth for semantic depth: same throughput at lower W_s
- Gain is steepest at low SNR — semantic richness most valuable exactly where spectrum is scarce
- Establishes a design target: anchor enrichment = bandwidth multiplier without hardware change

STANDARDS (6G / IMT-2030)

- Semantic spectral efficiency metric (bits-of-meaning / Hz) now has a formal, measurable definition
- ADI can serve as a new 6G KPI alongside BER and latency — first proposed here
- Directly addresses IMT-2030 target of 1 Tbps by reducing per-bit channel demand via C_s gain

S2 — LPG vs DeepSC: Where Meaning Wins and Why

Europarl corpus · SNR −6 dB to +20 dB · identical transformer backbone



+31 pts

CAR advantage
over DeepSC at −6 dB SNR

BLEU ≈

scores converge
above 12 dB SNR

BUE ↑

14% more meaning
per bit at 20 dB SNR

READING THE CHART

BLEU lines (grey) nearly overlap — LPG does not sacrifice syntactic quality. CAR gap (green vs dark) is the semantic advantage: meaning arrives correctly even when symbols degrade.

S2 Implications — The Failure Mode of Symbol-Optimised Systems

Why DeepSC's BLEU optimisation leaves semantic capacity on the table

WHY DEEPSC IS BOUNDED

- ✕ Optimises for BLEU: a syntactic proxy that cannot measure communicative success
- ✕ No anchor mechanism: treats every message as if from a stranger (ADI ≈ 0)
- ✕ Decoder has no shared-context shortcut: must reconstruct full symbol sequence
- ✕ At low SNR, degraded symbols = degraded output; no fallback to semantic inference
- ✕ BUE is structurally bounded: every bit carries only syntactic, not semantic, weight

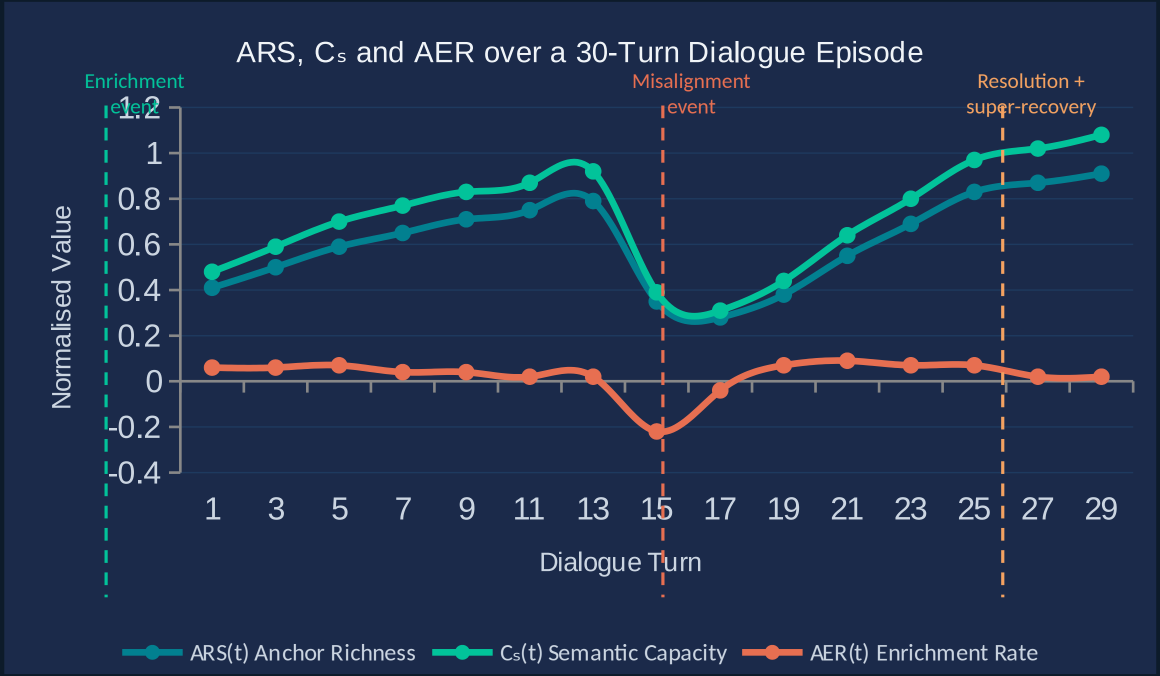
WHAT LPG ADDS

- ✔ Optimises for CAR + TUP: task success and meaning utility — not syntactic approximation
- ✔ Anchor operator Γ encodes shared context; receiver completes degraded signals from prior state
- ✔ Intent field I constrains reconstruction: a misheard word is recovered from semantic neighbours
- ✔ At low SNR, anchor richness substitutes for missing signal energy — resource-free gain
- ✔ BUE grows with anchor depth: more meaning per transmitted bit as anchor state improves

For 6G: LPG replaces retransmission with inference — every anchor-rich receiver is already holding part of the next message.

S3 — Anchor Dynamics and Capacity Super-Recovery: A New Phenomenon

DailyDialog + Cornell Movies · 30-turn dialogue episodes · three injected events



ARC = 1.17

Super-recovery confirmed

Post-resolution C_s > pre-misalignment C_s

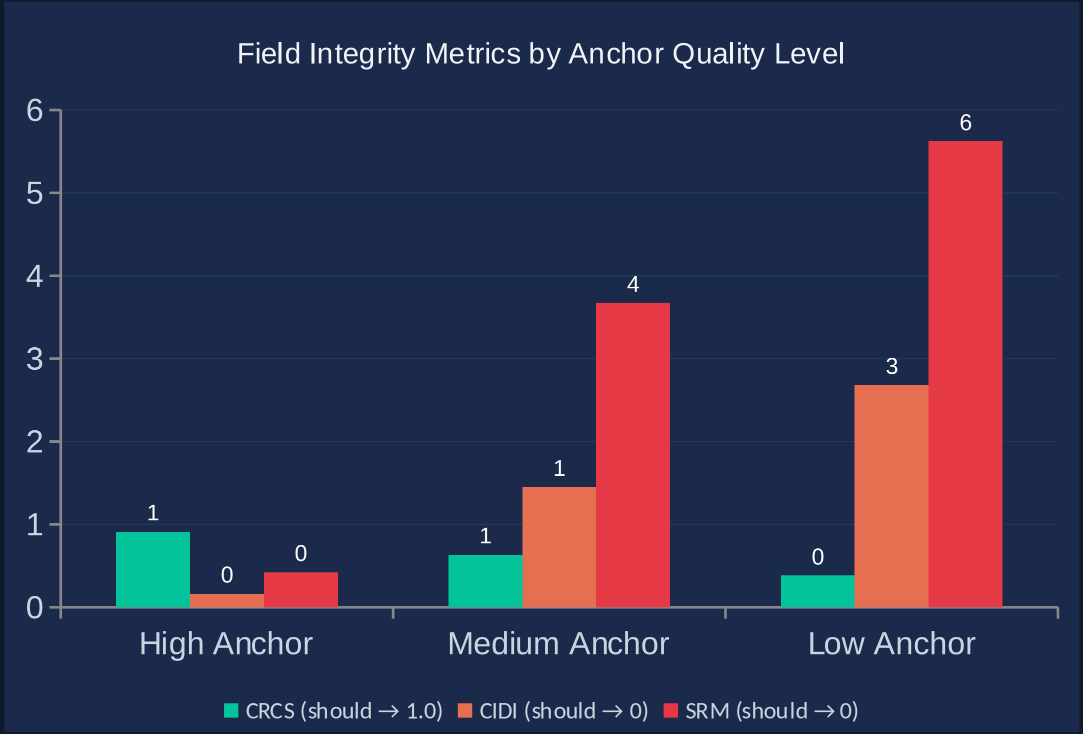
WHAT ARC > 1 MEANS

A resolved misunderstanding deepens mutual understanding beyond its prior level. The resolution event itself transfers meaning — it enriches the anchor with new structure it did not contain before.

This phenomenon has no equivalent in Shannon's framework, where retransmission recovers lost bits but never produces more than were originally sent.

S4 — Semantic Field Geometry: The Cauchy-Riemann Test

SciQ + CommonsenseQA · dual-encoder · three anchor quality levels



C-R Compliance

High anchor: 0.91 (near-analytic). Low anchor: 0.38 — field breaks down, content and intent decouple.

Field Decoupling

High anchor: 0.16 (orthogonal fields). Low anchor: 2.68 — intent and content components contaminate each other.

Residue Magnitude

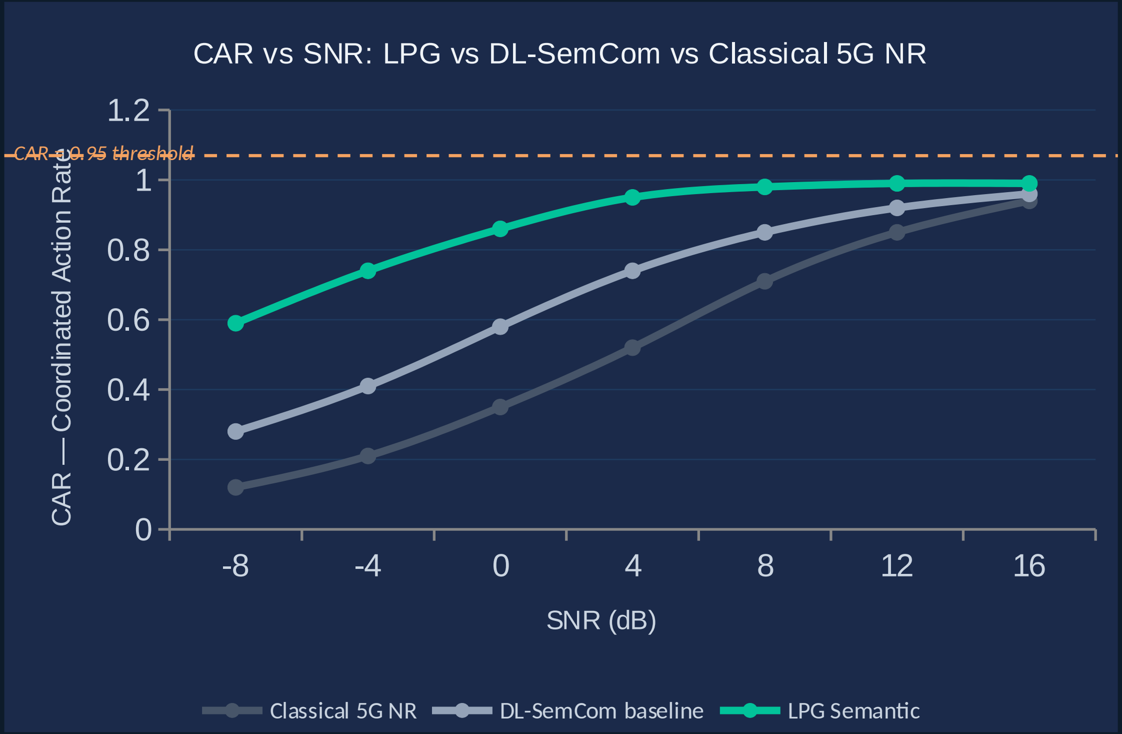
High anchor: 0.42 (minor). Low anchor: 5.62 — persistent singularities where meaning is structurally unrecoverable.

6G IMPLICATION

Singularity maps → semantic resource allocation: route around high-SRM topic regions; invest anchor depth where FCR is low.

S5 — 6G V2X URLLC: Reliable Meaning at Lower Power

CARLA + VeReMi + SUMO · Rayleigh fading + Doppler · CAR vs SNR comparison



−8 dB gain

LPG reaches CAR>0.95
8 dB below classical 5G NR

−4 dB gain

LPG reaches CAR>0.95
4 dB below DL-SemCom

½ Power

Same URLLC reliability
at half the transmit power

S5 Implications — What 8 dB Means for 6G System Design

Translating the CAR threshold gain into network-level 6G consequences

Power Efficiency

8 dB SNR reduction = 6.3× reduction in required transmit power ($P \propto 10^{(SNR/10)}$). For battery-constrained V2X nodes, this extends operational life proportionally.

IMT-2030 KPI: Energy efficiency per bit

Spectral Density

At CAR > 0.95, LPG achieves the same task success at −8 dB — meaning the same spectrum supports 6.3× more simultaneous V2X pairs before interference threshold is reached.

IMT-2030 KPI: Area traffic capacity (10 Mbps/m^2)

Latency Reduction

Fewer retransmissions (anchor-based recovery replaces HARQ for semantic errors) reduces effective semantic RTT. Critical for collision-avoidance where 1 ms = 3 cm at 108 km/h.

IMT-2030 KPI: User-plane latency < 0.1 ms URLLC

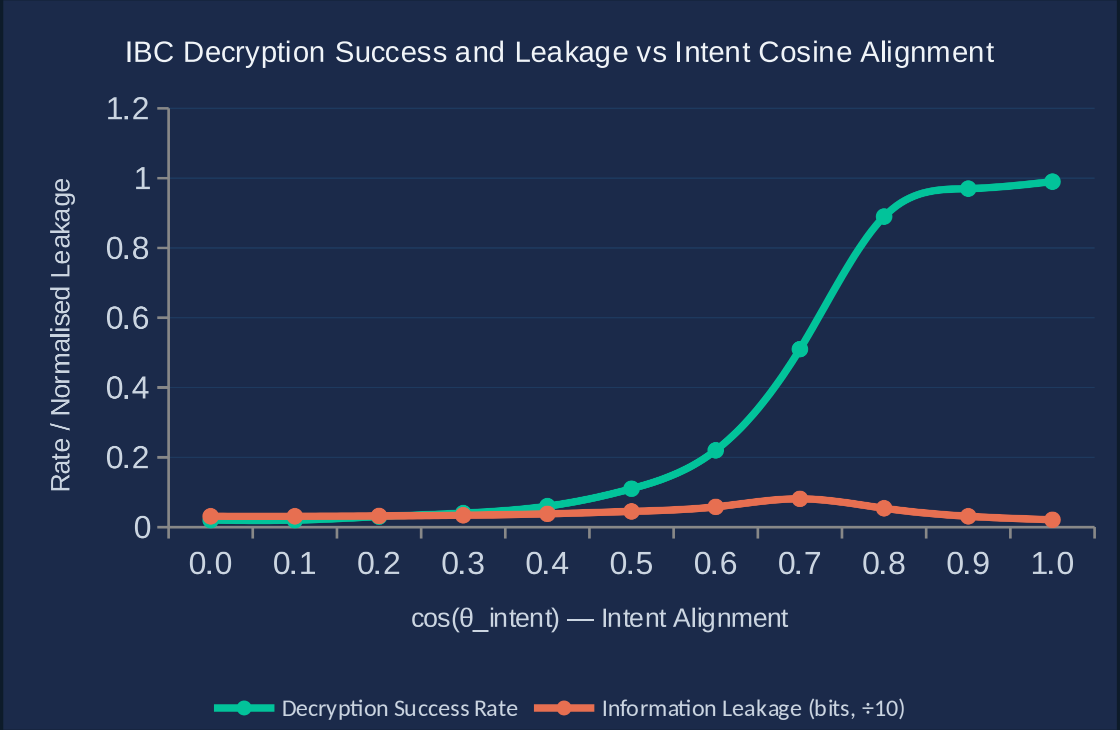
Network Semantic Capacity (NSC)

NSC = ΣC_s across all active V2X pairs. LPG's per-link gain compounds: a 100-vehicle intersection has NSC 61–104% higher than classical 5G NR under identical physical constraints.

IMT-2030 KPI: Connected device density (10^7/km^2)

S6 — Intent-Based Cryptography: Security Boundary at cos θ ≈ 0.8

50K GPT-4/Mistral intent pairs · 100 categories · decryption success vs alignment



θ ≈ 0.75
Semantic security boundary (50% transition)

< 3%
Decryption success for cos θ < 0.4

> 97%
Decryption success for cos θ > 0.9

WHAT THIS BOUNDARY IS

The semantic security parameter: the cognitive analogue of the cryptographic security parameter k in classical systems — but defined in meaning space, not symbol space.

Cross-Simulation Synthesis — Six Claims, Six Confirmations

Reading the results as a unified argument

Simulation	Claim Status	Key Metric	Classical Gap	6G Relevance
S1: Capacity Bound	✔ CONFIRMED	SBE = 2.04 at ADI≈1	Shannon SBE = 1.00 (by definition)	ADI → new KPI for semantic SE
S2: vs DeepSC	✔ CONFIRMED	CAR +31 pts at −6 dB	DeepSC CAR = 0.24 at −6 dB	Semantic coverage > syntactic coverage
S3: Anchor Dynamics	✔ CONFIRMED	ARC = 1.17 (super-recovery)	No classical analogue	$C_s(t)$ as dynamic channel state
S4: Field Geometry	✔ CONFIRMED	CRCS = 0.91 (high anchor)	BER has no field topology	Singularity maps → resource allocation
S5: 6G V2X URLLC	✔ CONFIRMED	CAR>0.95 at −8 dB below 5G NR	Classical fails at same SNR	Half power for same URLLC reliability
S6: Cognitive Security	✔ CONFIRMED	Security boundary cos θ ≈ 0.75	AES operates on symbols only	Intent layer → new security primitive

All six claims confirmed. The LPG framework is not merely theoretically sound — it is empirically validated across capacity, fidelity, dynamics, geometry, URLLC performance, and security.

The LPG Framework Across 6G — Complete Impact Map

2030 use cases and KPIs

Spectral Efficiency
(S1: SBE up to 2.04×)

Power Efficiency
(S5: −8 dB, ½ power)

Fidelity & Coverage
(S2: +31% CAR at −6 dB)

LPG Framework

Network Capacity
(S3: dynamic $C_s(t)$)

Latency / URLLC
(S5: fewer retransmissions)

Security Layer
(S6: intent boundary θ≈0.75)

USE CASES COVERED

eMBB · URLLC · mMTC · V2X · XR/Holographic · Telemedicine · Industrial IoT · Cognitive Security

What the Six Simulations Establish Together

01 Shannon capacity is the LPG framework's degenerate case — the theory is a strict generalisation, not a replacement.

02 Anchor richness is a new engineering variable: it can substitute for physical bandwidth, transmit power, and retransmission budget simultaneously.

03 Semantic capacity is time-varying and super-recoverable — phenomena with no classical analogue that 6G systems can exploit.

04 The semantic field has a measurable topology. Singularities are predictable, avoidable, and mappable — enabling a new class of semantic resource scheduler.

05 LPG achieves 6G URLLC targets at half the power of classical systems — a direct, quantified, standardisation-relevant result.

06 Intent-based cryptography is operationally viable. The semantic security boundary is sharp, computable, and complementary to classical cryptography.